AIRTEL SYNC · YTL HACKATHON 2026 · ENTERPRISE TRACK A

Udyam Digital

From 5,000 leads to a live connection, on a phone.
Approach, competitive analysis and impact.

7.4% → 11% lead-to-activation, baseline → pilot target	**17.6 → 8** days from lead to live	**192 / 192** automated tests pass	**0** ineligible bundles (today 15% break rules)

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2. Approach: problem, evidence, product decision, build
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Accel Business is the fictional stand-in for Airtel Business used in the case. Project numbers are computed from the supplied dataset (smb_leads.csv, product_catalogue.csv, mock GSTIN and Udyam registries, 47 KYC images) and the project's own analysis and test runs. Market and competitor facts are cited to public sources on the last page. Interactive version: the accompanying website.

1 · EXECUTIVE SUMMARY

A phone-first journey where rules decide, AI explains, and people handle exceptions.

Small businesses buying connectivity from Accel today wait for a call, make time for a visit, receive a slow quote full of product codes, collect paper documents and then chase installation. Only 7.4% of leads become customers, after 17.6 days and 3 status calls each.

What we built

- An AI advisor (text or voice, 8 languages) that asks at most 6 questions and returns a **rules-checked** bundle in plain words.
- An instant quote with add-on toggles and dependency locks: one monthly price.
- Smart KYC: DPDP consent first, live GSTIN/Udyam lookups, OCR on 4 documents and 13 cross-checks with specific fixes.
- A live tracker with expected dates, WhatsApp updates and automatic escalation to a named, language-matched agent.
- An admin dashboard: propensity-ranked leads, Call with outcome, and the 5 pilot metrics computed live.

Why it wins

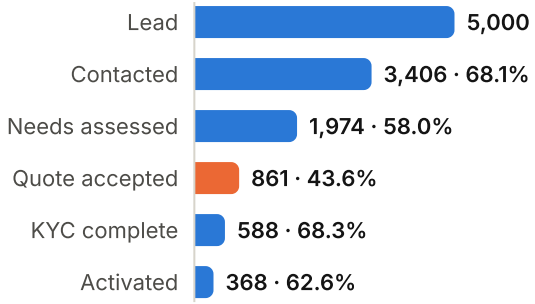
- **Aimed at the data.** 41.2% of drops are about reach and speed; the quote passes only 43.6%. The journey removes the visit and the wait.
- **Proven, not promised.** 192/192 tests, 15/15 judge-style scenarios, 12/12 KYC verdicts, a clean phone run from language to live.
- **Different from rivals.** Jio, Vi, Tata Tele and BSNL sell to MSMEs, but from public information buying still runs through a callback, an assessment or a booking-then-wait. Nobody combines advice, an eligible quote, KYC and tracking in one sitting.
- **Measurable.** Pilot targets: 11% conversion, 8 days, Rs 8,000 CAC, 11% churn, under 0.5 chase calls. At one quarter's 5,000 leads that is about 180 more customers and Rs 11.7 lakh less acquisition spend.

Recommendation

Launch the self-serve journey with **professional offices** (11.1% conversion, lowest CAC at Rs 10,695, 86% have a GSTIN, 5-day installs) and run **multi-site businesses** as an assisted track from day one. Measure the five metrics weekly against the Apr–Jun 2026 baseline, which the dashboard already reproduces exactly.

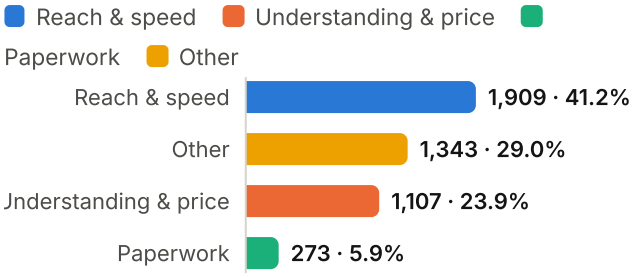
Start from the evidence: two-thirds of losses happen before anyone sees a price.

The funnel (5,000 leads, Apr–Jun 2026)



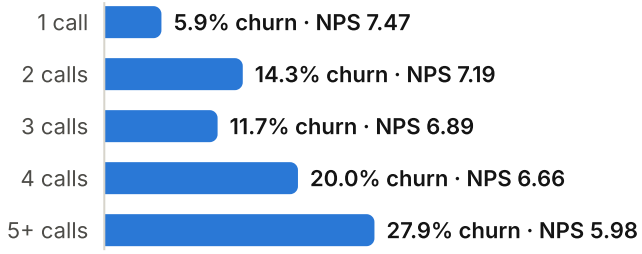
> The largest absolute loss is first contact (1,594); the lowest pass rate is the quote (43.6%, 32% for micro).

Why 4,632 leads dropped



> Reach and speed cause 41.2% of drops; paperwork only 5.9%.

Chasing status predicts churn



> Churn rises from 5.9% at one chase call to 27.9% at five or more: the tracker is a retention feature.

Other findings that shaped the design

- **Mis-selling:** 56 of 368 customers (15%) bought bundles the catalogue forbids.
- **GSTIN matters:** 9.6% conversion with a GSTIN vs 2.0% without; it gates 10 of 25 products, so the advisor asks for it at question 3.
- **Digital leads go unanswered:** 45% of website-form and 59% of WhatsApp leads are never contacted.
- **Lead scoring works:** the top 20% of scored leads hold 43.2% of activations (High bucket converts 15.9%).
- **KYC needs a safety net:** 6 of 12 sample applicants had planted problems; 2 need a person.

The product decision: a hybrid of AI and people.

CRITERION	WEIGHT	A · AI-FIRST SELF-SERVE	B · HYBRID AI + HUMAN	C · AGENT TOOL
Fit to the case (phone, language, field-force role)	25%	4	5	2
Addresses the observed drop causes	25%	3	5	3
Feasible within the hackathon	20%	4	3	4
Business value (CAC, conversion, churn)	15%	4	5	3
Risk (dead-ends, compliance, adoption)	15%	2	4	3
Weighted total (out of 5)		**3.45**	**4.45**	**2.95**

Scores are the team's judgement from the Phase 1 evidence. A dead-ends on exceptions (half the KYC sample needed a fix or review); C keeps the visit that causes 827 drops and the Rs 20–24k channel cost.

Design principles

1. **Rules decide, AI explains.** Eligibility, quantities and prices come only from deterministic code; LLM output naming any unlisted product or price is discarded.
2. **AI for acquisition, humans for exceptions.** KYC judgement calls, late steps, a stuck advisor or "talk to a person" go to a named agent with full context.
3. **Consent before KYC, minimum data, masked identifiers** (DPDP Act 2023).
4. **The customer always sees their own status:** expected dates from catalogue install days.
5. **Everything writes an event,** so the pilot metrics are computed from the database.

Build: six phases, each ending in automated tests and a phone-screen browser run

PHASE	WHAT WAS BUILT	WHAT IT PROVED
1 · Core advisor	OTP login, 8 languages, rules engine with reasons, 6-question advisor, guarded LLM explanations, recommendation card	10/10 profiles valid; 31 rule checks; no product codes on screen
2 · Instant quote	Pricing rule (micro at minimum, others at midpoint), toggles, steppers, dependency locks, "vs what you pay now", accept → order	Rs 1,597 and Rs 6,055 reference bundles match Phase 1
3 · Smart KYC	Versioned consent, GSTIN checksum and registry, Udyam lookup, OCR with deskew and blur checks, 13 cross-checks, verdicts that move the order	12/12 verdicts match planted problems
4 · Tracker & escalation	State machine, automatic install booking, SLA job, least-loaded language-matched assignment, agent console, simulator	Late orders escalate unaided; callbacks inside working hours
5 · Differentiators	Voice (browser speech, Sarvam/Bhashini server adapters), WhatsApp updates on 7 events + 24 h reminder, propensity lead dashboard with Call, live metrics strip	Full Hindi voice chat; spoken replies ≤ 180 characters; baselines reproduced live
6 · Hardening	15 judge-style scenarios, unit tests for pricing, state machine, escalation, mobile end-to-end run, security pass, deploy configs	192/192 tests; 0 page errors on the phone run

3 · RESULTS

What is proven today.

192 / 192
automated tests pass

15 / 15
judge-style advisor scenarios

12 / 12
KYC verdicts correct

0
page errors, language → live on a phone

15 advisor scenarios, typed the way a judge would

SCENARIO	LANG	BUNDLE	MONTHLY	REFUSED (WITH REASON)
tier2 kirana no gstin	en	Dukaan Digital	Rs 2,196	WhatsApp Business API Starter
five site restaurant chain	en	Chain Connect	Rs 19,766	—
one person salon	en	Dukaan Digital	Rs 1,897	—
manufacturer wants iot	en	Factory Line	Rs 19,919	—
small factory iot ineligible	en	Factory Line	Rs 6,017	Machine Monitoring IoT
two shops want sdwan	en	Chain Connect	Rs 7,590	Managed SD-WAN
hindi medical store	hi	Dukaan Digital	Rs 2,196	—
cancelled gstin wants whatsapp	en	Dukaan Digital	Rs 1,796	WhatsApp Business API Starter
ca firm suspended wants tollfree	en	Office Pro	Rs 6,055	Toll-free Number
tiny office wants telephony	en	Office Pro	Rs 4,170	Cloud Telephony / SIP
coaching centre tier1	en	Office Pro	Rs 9,805	—
messy caps and typos	en	Office Pro	Rs 6,055	—
tamil chips textile unit	ta	Factory Line	Rs 16,675	—
marathi bakery add gst later	mr	Dukaan Digital	Rs 2,196	WhatsApp Business API Starter
gujarati three pharmacies	gu	Chain Connect	Rs 12,733	—

Smart KYC on the 12 sample applicants

K01 ✓ Blocked	**K02** ✓ Verified	**K03** ✓ Verified	**K04** ✓ Re-upload needed
K05 ✓ Re-upload needed	**K06** ✓ Human review	**K07** ✓ Human review	**K08** ✓ Verified
K09 ✓ Verified	**K10** ✓ Verified	**K11** ✓ Re-upload needed	**K12** ✓ Verified

✓ = verdict matches the planted problem. 6 verified, 3 re-upload, 2 human review, 1 blocked.

Security and data protection

- Every log line masks GSTIN, Udyam, PAN and phone numbers, including request URLs.
- Documents encrypted per file, stored outside the web root, purged 90 days after verification (30 if abandoned).
- Rate limits: OTP 3 per phone and 20 per IP per 10 minutes, 5 wrong codes; registry lookups 20 per user per 10 minutes.
- Secrets required from the environment outside development; the API refuses to start without them.

Everyone sells to small businesses. Nobody sells the whole journey.

8.84 crore
enterprises on the Udyam portals, 30 Jun 2026 [1]

12%
of MSMEs fully digitally mature (DMI 58.0) [2]

< 3%
SME adoption of enterprise digital services vs 75–90% for large firms [3]

INR 954 bn
enterprise telecom opportunity through FY2030 [4]

The market is large, formalising fast and still mostly undigitised, which is why every Indian operator now runs an MSME programme. The question is how a business actually buys. From public information, the purchase still runs through a callback, an assessment or a booking followed by a wait. Verification vendors sell excellent components, not journeys.

Capability comparison

CAPABILITY	ACCEL TODAY	JIOBUSINESS	VI BUSINESS	TATA TELE	BSNL / ACT	KYC VENDORS	UDYAM DIGITAL
Self-serve purchase, start to finish	○	◐	○	○	◐	○	●
Guided discovery that ends in a bundle	◐	○	◐	◌	○	○	●
Eligibility rules enforced at sale	○	◌	◌	◌	◌	○	●
Instant, transparent quote with add-ons	○	◐	○	◌	◐	○	●
Digital KYC with GSTIN / Udyam checks	○	◌	◌	◌	◐	●	●
Live order tracker with expected dates	○	◐	◌	◌	◐	○	●
Vernacular voice in the buying journey	○	◌	◐	◐	○	○	●
Named human on exceptions, full context	◐	◐	◐	◐	○	○	●
Lead scoring for the sales team	○	◌	◌	◌	◌	○	●

● yes, publicly evidenced · ◐ partly · ○ not offered · ◌ not public. Assessed from public sources, not from inside each product, so competitor ratings may understate unpublicised features.

Competitor profiles

JioBusiness HIGH THREAT ON PRICE

Offer. Fibre plus productivity tools; seven plans from Rs 901 to Rs 10,001 at launch [5], now Business Broadband Max plans from Rs 801 with JioMeet and attendance tools [6].

How SMBs buy. Register interest and wait for an executive [7]; the solution page still leads with "Want a callback?", and the app manages accounts after purchase [8].

Gap. No public evidence of guided, eligibility-checked buying or digital KYC in the purchase.

Our edge. Purchase and KYC finish in one session; the price is transparent and rules-checked.

Vi Business ADVISORY LEADER

Offer. Ready for Next: a free 10-minute digital maturity assessment with Dun & Bradstreet [9]; 1.9 lakh MSMEs assessed by Feb 2025 [10]; state MoUs with local-language training [11][12].

How SMBs buy. The assessment recommends areas; purchase goes through sales offers.

Gap. Diagnoses needs without turning them into a priced, eligible bundle.

Our edge. Six questions end in a plan with one monthly price, and the owner can buy it immediately.

Tata Tele Business Services CLOSEST STRATEGIC RIVAL

Offer. 100% SMB-focused [13]; Smartflo cloud communications and a WhatsApp Business platform [14].

Direction. July 2026: a unified AI stack with Tata Communications, including voice agents with their own numbers and a digital B2B exchange to discover and adopt services [15].

Gap. Announced, not yet an end-to-end onboarding journey with KYC and tracking.

Our edge. A working journey today; the advisor's voice layer already speaks 8 languages.

BSNL & ACT Fibernet LOW THREAT ON EXPERIENCE

Offer. BSNL takes bookings online or via WhatsApp "Hi", with document upload; an engineer calls within 3–7 days [16][17]. ACT sells city-specific SME plans with installation charges [18].

Gap. Connectivity plans, not advised bundles; no expected dates or proactive updates in public material.

Our edge. Bundle advice, instant quote, expected dates and a live tracker.

IDfy · Signzy · HyperVerge PARTNERS, NOT RIVALS

Offer. GSTN and Udyam verification, OCR and video KYC as APIs [19][20][21].

Our position. KYC checks sit behind one interface, so a vendor API can replace the mock registry without changing the customer journey.

The status quo THE REAL COMPETITOR

A visit, a hand-made quote, paper KYC and 17.6 days; 15% of sold bundles break catalogue rules; customers chase status 3 times.

Our edge. Every leak in the Phase 1 data has a feature aimed at it and a metric to prove it.

Where Udyam Digital sits

Positioning

	ADVICE BEFORE BUYING	HOW THE PURCHASE HAPPENS	AFTER PURCHASE
Operators (Jio, Vi, Tata Tele)	Plans pages; Vi's 10-minute maturity assessment	Callback, sales executive or partner	Self-care apps and portals
Fixed-line ISPs (BSNL, ACT)	City plan lists	Online booking, then an engineer visit	Self-care portal
KYC vendors	—	APIs a telco wires into its own flow	—
Udyam Digital	6-question advisor, eligible bundle with reasons	Instant quote, consent, KYC checked on upload, in one session	Live tracker, WhatsApp updates, named agent when late

How competitors could respond, and our defence

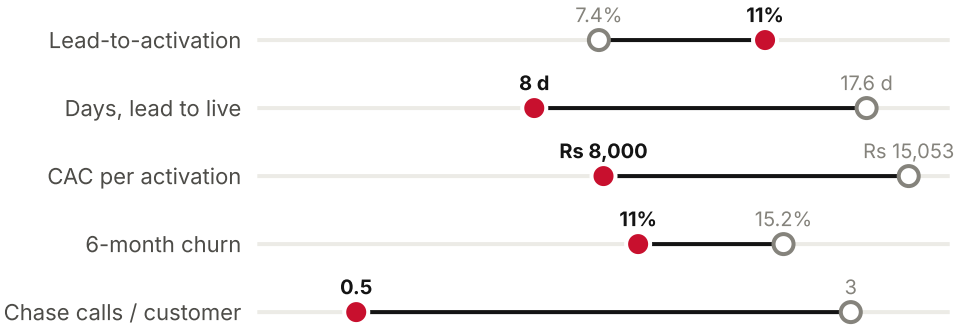
- **Price cuts (Jio).** We don't compete on discounts: the data shows speed and clarity drive more drops (41.2%) than price (23.9%), and micro customers already get catalogue-minimum pricing.
- **AI agents (Tata).** Our moat is the deterministic rules engine and the KYC + escalation plumbing behind the conversation, which an agent alone doesn't provide.
- **Advisory reach (Vi).** Route assessment-style traffic straight into the advisor so diagnosis ends in a purchase.

5 · IMPACT

What a 6-month pilot could change.

Targets come from the data: professional offices already convert at 11.1%, referrals at 14.8%, and core installs take 5 days. They are pilot proposals, not results.

The five pilot metrics: baseline (Apr–Jun 2026) → 6-month target



What the targets mean in numbers

PER QUARTER	TODAY, 5,000 LEADS	PILOT TARGETS, 5,000 LEADS	PILOT TARGETS, 50,000 LEADS
Activations	370	550 (+180)	5,500 (+1,800)
Acquisition spend	Rs 55.7 lakh	Rs 44.0 lakh	Rs 4.40 crore (vs Rs 5.57 crore)
Status-chase calls	1,110	275	2,750 (vs 11,100)
Extra customers still paying at 6 months	—	+176	+1,757
Added yearly revenue from them (× Rs 6,900 median bundle)	—	Rs 1.46 crore	Rs 14.55 crore

Arithmetic on the stated targets and the dataset's baselines (7.4%, Rs 15,053, 3.0 calls, 15.2% churn, Rs 6,900 median bundle). The website has an interactive calculator for other assumptions.

PROVEN NOW

- Zero ineligible bundles: checked by 30 rule tests and 25 advisor scenarios.
- One-session journey on a phone, language to live, with real document uploads.
- Bad documents caught at upload with a specific fix.
- Late steps escalate unaided to the right agent.
- Baselines reproduce exactly from the database.

TO VALIDATE IN THE PILOT

- Share of drops recovered: between 32% (directly addressable) and 69% (including partly addressable).
- A 50/50 self-serve / assisted mix with assisted deals 40% cheaper.
- KYC completion from 68.3% to 85%; first response to escalations within 4 business hours.
- Whether vernacular voice closes the quote gap (40% regional vs 45%).

Impact beyond the numbers, and what happens next

Impact beyond the numbers

- **For owners:** no visit, no waiting for a quote, one clear price in their language, and a date they can check instead of calling.
- **For Accel:** field staff and partners spend time on multi-site deals, KYC exceptions and High-propensity leads, with credit for app-completed sales.
- **For compliance:** consent, masking, encryption and retention are built in, and every eligibility decision is reproducible.

Next steps

1. Deploy the pilot (frontend on Vercel, API on Render, seeded SQLite) and route website and WhatsApp leads into the advisor.
2. Run 6 months with professional offices; multi-site via account managers.
3. Swap mock registries for GSTN/Udyam vendor APIs and connect Twilio WhatsApp and a speech provider.
4. Review the five metrics weekly by segment and channel; re-score leads monthly.

Sources

Project figures: supplied hackathon dataset and the project's analysis, test and end-to-end outputs (docs/kyc_results.md, docs/advisor_scenarios.md, the 192-test suite). Market and competitor facts:

1. KNN India: over 8.84 crore MSMEs registered on Udyam platforms (Jul 2026)
2. CRN India: Vi Business Ready for Next MSME study 2025 (DMI 58.0, 12% fully mature)
3. Multibagg: India's SMEs under-3% enterprise digital services adoption (Jio filing)
4. Frost & Sullivan: INR 954 billion enterprise telecom opportunity through FY2030
5. Disruptive.Asia: Jio launches seven plans, Rs 901 to Rs 10,001
6. TelecomTalk: Jio Business Broadband Max 801 plan (Mar 2026)
7. CIO Tech Outlook: register interest and wait for a JioBusiness executive
8. Jio: JioBusiness Solution page (callback form, self-care portal)
9. Vi Business: Ready for Next assessment (10 minutes, free)
10. Vi press release: 1.9 lakh MSMEs assessed by Feb 2025
11. TelecomTalk: Vi MoU with West Bengal, local-language advisory
12. TelecomTalk: Vi Business MoU with Andhra Pradesh (Jul 2026)
13. TelecomTalk interview: TTBS is 100% focused on SMBs
14. TTBS: end-to-end digital solutions, Smartflo, WhatsApp platform
15. NSE filing: Tata Communications and TTBS unified AI stack for SMBs (Jul 2026)
16. BSNL: book FTTH online or WhatsApp 'Hi' to 18004444
17. CitizenNest: BSNL online application, engineer contacts in 3–7 days
18. ACT Fibernet: business broadband plans
19. HyperVerge: GSTN verification
20. G2: IDfy KYC, KYB and 140+ verification APIs
21. Signzy KYB API: GSTIN, CIN, PAN verification